ALGEBRAIC SPACES WITH A DIMENSION FUNCTION ARE EXACTLY FINITE GROUPOID QUOTIENTS OF SCHEMES WITH DIMENSION FUNCTIONS

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Editorial note. The groupoid dimension-function input is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Finite flat equivalence global. Properties of Spaces, proposition finite-flat-equivalence-global (cross-chapter, named). The quotient structure.
- Dimension functions on groupoids. lemma-dimension-function-groupoid (same chapter, NOT carded, declared external). Dimension functions on groupoid quotients.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02JW: focused-seeds/geometry/S9-decent-scheme-with-dimension-function/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NM: focused-seeds/geometry/S9-decent-scheme-with-dimension-function/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02I9: focused-seeds/geometry/S9-decent-scheme-with-dimension-function/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a locally Noetherian and universally catenary scheme. Let $\delta : S \to \mathbf{Z}$ be a dimension function. Let X be a decent algebraic space over S such that the structure morphism $X \to S$ is locally of finite type. Let $\delta_X : |X| \to \mathbf{Z}$ be the map sending x to $\delta(f(x))$ plus the transcendence degree of $x/f(x)$. Then δ_X is a dimension function on $|X|$.*

Proof. Let $\varphi : U \to X$ be a surjective étale morphism where U is a scheme. Then the similarly defined function δ_U is a dimension function on U by Morphisms, Lemma 02JW. On the other hand, by the definition of relative transcendence degree in (Morphisms of Spaces, Definition 04NM) we see that $\delta_U(u) = \delta_X(\varphi(u))$.

Let $x \rightsquigarrow x'$ be a specialization of points in $|X|$. by Lemma 03IL we can find a specialization $u \rightsquigarrow u'$ of points of U with $\varphi(u) = x$ and $\varphi(u') = x'$. Moreover, we see that $x = x'$ if and only if $u = u'$, see Lemma 03K5. Thus the fact that δ_U is a dimension function implies that δ_X is a dimension function, see Topology, Definition 02I9. □